

Tablets for deeply disadvantaged older adults: Challenges in long-term care facilities

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ABSTRACT

There is a gap in the literature regarding the effects of ICT interventions on socially disadvantaged older adults that live below the poverty line and are institutionalized in long-term care facilities. Research suggests that technology might lead to a slowdown in cognitive deterioration, and could create opportunities for social connectedness, helping alleviate social loneliness and low self-esteem. This paper describes the experience and results of the implementation of a one laptop per elderly plan (*Ibirapitá* Plan) in a public geriatric hospital, within a highly vulnerable population, consisting of patients with dementia and low educational attainment, who are in general abandoned by their families, highly dependent, with many years of residency at the hospital and no previous usage of technology. We are the first to study the impact of a program of this kind directed to such a deeply disadvantaged population. We assess the effects on patients' cognitive deterioration, loneliness and self-esteem. Results suggest no significant effects of the tablets intervention on these variables. Our findings are important in the avoidance of techno-optimism among policy makers and the understanding of mechanisms that may prevent positive results from ICT programs for the elderly.

1. Introduction

There has been hope that the use of technology by the elderly might lead to a slowdown in cognitive deterioration and an improvement in their feelings of loneliness, self-esteem or depression. The appearance of tablets is seen as especially promising due to characteristics such as their cheap price, user-friendly interface, touch-screen and lightness. But recent studies have challenged the techno-optimistic interventions that ignore the contextual negotiations that seniors engage with while using computers (Castleton et al., 2020; Duarte and Coelho, 2019; Righi et al., 2017).

Previous literature has investigated how engaging older adults with a new mentally challenging activity involving technology has cognitive or socioemotional effects. Most programs studied in the literature focus on older adults from developed countries that are free of neurological and psychiatric conditions, and are not living under poverty conditions nor are institutionalized.

There is a gap in the literature regarding the effects of tablet computers on socially disadvantaged older adults that live below the poverty line and are institutionalized in long-term care facilities. Our main contribution is that we have exploited a unique opportunity for

exploring this topic. Uruguay is a developing Latin American country that experiences a fast aging population (OPP, 2018). Uruguay is deploying a national plan to provide each older adult in the country with a tablet. This plan is called the *Ibirapitá* Plan ("Plan Ibirapitá" in Spanish).

In this framework, we took the opportunity to study the impact of the use of tablets in an especially vulnerable population: the patients of the *Piñeyro del Campo* Hospital ("Hospital Dr. Luis Piñeyro del Campo" in Spanish), a public hospital that is focused on deeply vulnerable elderly located in Montevideo, the capital of Uruguay. Recent studies suggest that these technology interventions might lead to a slowdown in cognitive deterioration, and could create opportunities for social connectedness, helping alleviate social loneliness and low self-esteem. Employing a field experiment and a difference-in-difference approach, we assess the effects of this tablet program on patients' cognitive skills and connectedness. The follow-up round –i.e. the further observation of key variables on treatment and control seniors– was conducted 9 months after the start of the program. The assessment included observational data gathered by the hospital (dementia, cognitive and antidepressant medication, workshop attendance and external contact), results of tests administered by the hospital (Pfeiffer test (Pfeiffer, 1975), Clock drawing test (Shulman et al., 1986)) and tests

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administered by volunteers (self-esteem test, loneliness test).

2. Previous literature

Righi et al. (2017) addressed a number of challenges human-computer interaction designers and researchers deal with when designing digital technologies for older people. They conducted a 5-year research-through-design study, which combined ethnography, participatory design and real-life evaluation, to explore the design and use of technologies aimed to enhance the social life of older people. The paper explores widespread assumptions about the social category of older people and the meaning of technologies for them. They argue that technologies designed 'for older people' should be designed to meet situated and dynamic needs/interests of the communities (and not only of care) to which they belong. In the following sub-sections, we offer recent findings and discussions in the design of technologies for an ever-increasing aging population.

2.1. Tablets for the elders

Macedo (2017) offers a recent review of 41 empirical studies on ICT use and acceptance among older adults. The adoption of digital technologies by the elderly is strongly favored by facilitating conditions such as support through step-by-step guidance, providing a friendly space to use trial and error methods, or the provision of a manual (Barnard et al., 2013; Vaportzis et al., 2017; Chiu et al., 2016). Tablets are seen as a friendlier technology than personal computers for older adults. In Austria, a study revealed high acceptance of tablets and satisfaction rates among a group of 11 seniors (60+) (Werner et al., 2012).

According to Magsamen-Conrad et al. (2015) the use that older adults give to tablets is, in order of relevance: information seeking, amusement and killing time, relationship maintenance, style and organization. Vroman et al. (2015) show that technology is used to maintain family and social connections and to access information on health and routine activities. Coelho and Duarte (2016) present a survey of older adults' use of social network services and social applications, consolidating 31 papers from the previous ten years. The authors recognize that the main motivation for adopting these tools is the possibility of keeping in contact with family members.

The perceived benefits of technology use may be outweighed by the costs of such use. Negative aspects are related to technology creating inconveniences, unhelpful features, as well as security and reliability concerns (Mitzner et al., 2010). Previous research show mixed results about the effects of ICT programs on adults well-being:

2.2. Preventing cognitive deterioration and loneliness

2.2.1. Impact on cognitive features

Older adults who use computers tend to have higher levels of cognitive abilities (Wagner et al., 2010). In this vein, Whitlock et al. (2012) found that computer-based video games intended for cognitive training can be effective for improving attention on an intervention with 39 older adults (aged 60 to 77) from North Carolina, United States. It looks also promising the use of tablets for cognitive training for people with schizophrenia. Biagianti et al. (2017) made a study of 111 subjects (with a 36.6% attrition rate). They found significant progress in global cognition and verbal memory.

2.2.2. Fostering connectedness

Khosravi et al. (2016) reviewed 34 papers on the impact of technology on older adults' isolation. They report that, in general, the use of technology lowers the level of feelings of loneliness. A large-scale study (Cotten et al., 2012) showed a clear contribution of an ICT intervention to the mental well-being of retired older adults, reducing depression categorization by approximately 20–28%. The study involved 7839

retired Americans, 50 or more years old. Tsai et al. (2015) interviewed 21 participants from the United States, aged 69 to 91 years old, with a more than average socioeconomic status, on their use of tablets. The impacts on their lives were profound, helping them feel connected, preventing isolation. This result is sustained by the literature review presented in Wagner et al. (2010). Sinclair and Grieve (2017) studied usage of Facebook in a population of 280 participants aged 55 and older and concluded that while Social Network Services were originally created for the young (Ellison et al., 2007), older adults can also gain social connectedness through Facebook, and, moreover, to the same extent as their younger counterparts.

2.3. Negative or null effects on cognitive abilities and connectedness

2.3.1. Failure to develop cognitive skills

Chan et al. (2016) described a study on 54 older adults (age 60–90). Eighteen participants received a tablet and the rest were divided into two control groups. The cognitive functions evaluated were processing speed, mental control, episodic memory and visuospatial processing. Though they found improvements in some features, no difference in mental control or visuospatial processing between the groups was found. A similar intervention was held at Edinburgh, United Kingdom, with 48 subjects aged 65 to 76 (Vaportzis et al., 2017). Though the treated group showed greater improvements in processing speed compared with the controls, they did not differ in verbal comprehension, perceptual processing, or working memory.

2.3.2. Failure to foster connectedness

A focus group study with seventeen subjects in the United Kingdom (Hill et al., 2015), recognized that without appropriate skills, or measures to tackle the fear associated with technology use, many older adults perceived the risk of increased social isolation and reduced access to key services as more of society and business moves exclusively online. Similar conclusions were reached by Wright (2014) after a study of 52 members of a United Kingdom branch of the University of the Third Age. A study in Finland and Slovenia showed that not every participant of a computer training and subsequent ICT program experienced a reduction in loneliness: the effects depended on the older people's previous lifestyle (Blažun et al., 2012). Fokkema and Knipscheer (2007) found similar results.

2.4. Mechanisms

Much research has focused on the introduction of ICTs in older people, but it has generally understood them as external factors who act as a sort of independent variable that impacts the seniors' lives. There remains a dearth of empirical research into how aging and technology are co-constructed. Deploying qualitative ethnographic research comprised of semi-structured interviews, jottings, and field notes, Castleton et al. (2020) explored the adoption of new ICTs, and the corresponding actor-network, by seniors who have typically resisted such technology. Their findings suggest that not only was the establishment of the sociotechnical network rife with complexities, disruptions, and fissures, but that the user was heavily influenced by her self-perception of the potentialities of modern ICTs. In this vein, a study in Germany (Dahn et al., 2014), made recommendations for the design of the user interface and accompanying support measures. It evaluated the use of Android tablet computers by a group of 19 seniors, aged between 53 and 82 over three months. It recommended that seniors should be offered personal support when acquiring a tablet and that support measures need to be prepared to handle the heterogeneity of the group. Support is especially necessary at the beginning of device usage. In this vein, in Houston, Texas, United States, in 2001, a study suggested that technology itself does not lead to improvements in self-esteem or depression. Only when there was help from the nursing staff significant differences appeared (Billipp, 2001). The intervention involved 65-year

old and older computer-illiterate persons, exposed to interactive computer use.

In summary, we could extract the following characteristics from the literature review:

- a) Most studies focus on developed countries. We found no rigorous article describing experiences in Latin America or in Africa.
- b) Most studies involve highly educated subjects.
- c) There is no report on the impact of technology on highly vulnerable seniors.
- d) Experiences involve tens of subjects, with as few as 11, last for some months, and there is an important percentage of attrition in many experiences, because of either deaths or decreased interest.
- e) Not all studies suggest a positive impact of tablets on seniors' cognitive abilities and connectedness

3. The program: *Ibirapitá* Plan

Uruguay is a developing country with high ICT indicators. It shows a "very high" e-government index (["EGOVKB | United Nations > Home," 2018](#)): in fact, the highest in Latin America. It also reports the highest ITU ICT development index in Latin America, as well as the highest number of broadband subscriptions per capita ([ITU, 2017](#)).

Since 2007, Uruguay has deployed a One Laptop per Child program called *Ceibal Plan* ([Yanguas, 2020](#); [Kozma and Vota, 2014](#); [Pittaluga and Rivoir, 2012](#)). It succeeded in reducing the digital divide in Uruguayan childhood.

In 2015, the country began a novel initiative called the *Ibirapitá* Plan. This program promotes the inclusion of retirees from all over the country by delivering a tablet free of charge. The tablet is a touch-screen (8") mobile device that runs an Android operating system, with an interface specially developed to be intuitive and friendly for its recipients. The program conducts training workshops of 2-hours duration and provides the necessary support.

3.1. *Piñeyro del campo hospital*

The "*Hospital Dr. Luis Piñeyro del Campo*" (HPC) is a public geriatric center. It was founded in 1860 as "*Asilo de Mendigos*" (Beggar's Home).

Today it houses hundreds of elderly people in vulnerable situations. All of them show low-income background. Some have been assigned by a judge to stay there. In general, they lack an actively present family.

The structure of the hospital is made up of wards, which could be described as large blocks that communicate with each other through a large garden, or small park. [Fig. 1](#) shows a plan view of the hospital premises.

- Ward A is for residents with high dependency - specifically with high physical dependence -, and low mobility.
- Ward B is for residents with low dependency in general, they are the ones that are in better conditions, since they lack mental problems and are self-able in physical terms.
- Ward C is for residents who suffer from psychiatric problems.
- Ward D is for people who suffer dementia. For security reasons, preventing attempts to escape, the ward is metal fenced.

Each ward has a different structure, but they all have two floors. On the upper floor is the dining room and the females' bedrooms, and on the floor below the male bedrooms and a smaller dining room.

Finally, there is an occupational therapy ward, which is a space with books and computers, little used for these purposes, but where painting workshops are held. Some patients come only during the day to the hospital, but reside at their own homes, using the Day Center.

The hospital provides the following health benefits: prevention, evaluation and comprehensive geriatric treatment; psychiatric care; nutritional assessment and food planning; dental service, control and

treatment including dental prostheses; nursing care; physiotherapy and occupational therapy; physical education; recreational gymnastics; podiatric care, hairdressing and manicures; medication and diapers or other devices for incontinence.

It also offers intervention and stimulation programs: therapeutic horticulture, recreation group activity, workshops of cognitive stimulation; selection, adjudication and loan of walking aids and substitutes: canes, walkers, wheelchairs, etc. There are also religious services held.

4. Method

4.1. *Design and setting*

In order to identify the effect of tablets on cognitive outcomes (dementia, cognitive and antidepressant medication, Pfeiffer test, Clock drawing test) and connectedness outcomes (workshop attendance, external contact, self-esteem test, loneliness test), we exploited the fact that the number of tablets that the hospital received from the government (*Ibirapitá* Plan) - 60 tablets - is less than the number of elders selected by the hospital to participate in the program - 93 individuals. The elders that would participate in the program were selected by lottery. This allocation rule ensured that the group randomly assigned to participate in the tablet program is similar at baseline to the group of elders not drawn in the lottery. The Universidad de Montevideo coordinated the program in the hospital and was in charge of providing training to the participants.

4.2. *Participants*

[Table 1](#) shows the demographic data of the participants. At the beginning of the intervention the 93 participants were 78 years old average, 42% were males and 58% females. Most of them were low educated and had been at the hospital for an average of 5 years, although one participant had been at this hospital for more than 25 years. Most of them (71%) report no previous use of technology. A quarter of the subjects were nonresident patients of the hospital receiving daytime care. These subjects in general have higher autonomy and better cognitive status. Almost half of the sample reported no contact outside the hospital at baseline; their lives were restricted to what was experienced inside it.

One of the distinct aspects of the present study is the vulnerability of the subjects. [Table 2](#) shows some indicators related to the health status of the participants at the beginning of the intervention. There was a high percentage of dementia (45 percent of the sample suffered dementia), which were measured by professional diagnosis. Cognitive medication and antidepressant use were relatively common among the participants (20 percent and 49 percent respectively).

Vulnerability is evident through the mental health status of the subjects involved. It was assessed through two different tests: the Pfeiffer and Clock drawing tests. These tests were undertaken by the specialized professionals at the hospital.

The Clock drawing test assesses cognition focusing on executive function and praxis. It is quick to administer, easy to score, well tolerated and accepted by patients, and relatively independent of culture, language and education ([Shulman et al., 1986, 2000](#)). "Performance on this test depends on intact attention, receptive language, executive functions (e.g. planning, organization, simultaneous processing and self-monitoring), and visuo-construction skills" ([Shulman, 2000](#), p. 230). The Pfeiffer test consists in a brief questionnaire that shows good sensibility and specificity as an index. This test is cited as "a short portable mental status questionnaire for the assessment of organic brain deficit in elderly patients" ([Pfeiffer, 1975](#)) and evaluates both short and long term memory, information on daily incidents, calculating ability and orientation.

[Fig. 2](#) shows that forty-seven percent of the sample presented moderate to severe deficit (less than or equal to 5 points in the Clock

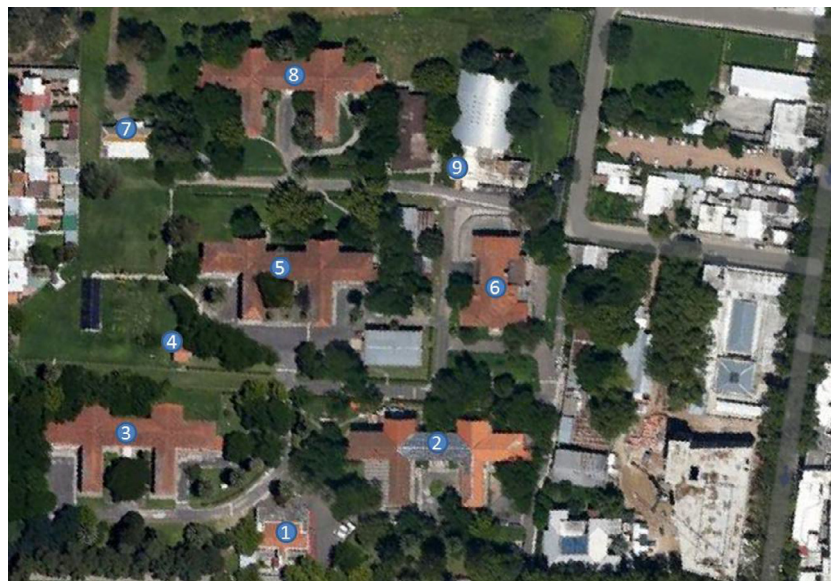


Fig. 1. Top view of Piñeyro del Campo Hospital. 1-Secretariat; 2-Ward A; 3-Ward B; 4-Security; 5-Ward C; 6- Assembly Hall; 7-Day Center; 8-Ward D; 9-Occupational therapy ward.

Table 1

Demographic data of participants at baseline.

Source: Survey at baseline conducted by the research team & administrative data provided by the hospital about the 93 individuals selected by the hospital authorities to participate in the project.

	#Obs.	Mean	SD	Min	Max
Age (in years)	91	78.08	7.54	63.00	100.00
Male	93	0.42	0.50	0.00	1.00
Education (in years)	92	5.77	3.61	0.00	16.00
Previous use of technology (1 = high; 0 = low)	92	0.29	0.46	0.00	1.00
Number of months in the hospital	93	59.70	66.33	0.00	303.00
Subjects with external contact	92	0.57	0.50	0.00	1.00
Not living at the hospital (only daytime attendance)	93	0.25	0.43	0.00	1.00

Table 2

Health status of participants at baseline.

Source: Administrative data provided by the hospital about the 93 individuals selected by the hospital authorities to participate in the project.

	#Obs.	Mean	SD	Min	Max
Dementia	93	0.45	0.50	0.00	1.00
Cognitive medication	93	0.20	0.41	0.00	1.00
Antidepressants medication	93	0.49	0.50	0.00	1.00
Pfeiffer test	91	3.27	3.13	0.00	10.00
Clock drawing test	66	5.00	2.92	0.00	10.00

drawing test scale). However, this number increases to 68% when taking into account cases of mild deficit (less than or equal to 7 points). On the other hand, the Pfeiffer test presented relative better results, but also confirmed the vulnerability of the population sample. Even adjusting the categorical scale with respect to the educational level of the subject (where a lower educational level receives more tolerance in Pfeiffer), the results do not vary.

4.3. Intervention

The tablet intervention in the *Piñeyro del Campo* Hospital started in December of 2016 with a short pilot with four residents, one from each ward. Each one received eight training sessions of one hour and a half.

The pilot outputted some knowledge on practices: the importance to consider the flaws of hospital's internet infrastructure; the understanding of the real interests of the elderly; the seniors' daily routines; the challenges in the interaction with the hospital staff.

In March 2017, volunteers from the Universidad de Montevideo started to train the individuals randomly assigned to the treatment group. The intervention was face-to-face and delivered by undergraduate students.

The research team trained the volunteers to provide support to such fragile people and about the appropriate way to approach them. A fulltime supervisor was hired by the research team to follow – daily – each volunteer and answer any question or solve any difficulty. Most difficulties that volunteers encountered were not related specifically with seniors but with the environment: problematic accessibility of tablets (placed in distant lockers because of security); irregular internet and Wi-Fi in the hospital; stairs that prevent the movements of seniors within the ward looking for better internet access; the competition from other networks such as television and radio that did not demand much from seniors. Negative identity might qualify as a problem arising in training time: some seniors stressed that the tablets were not for them but for the younger generation. Also, the difficulty in the understanding of basic concepts due to cognitive deterioration challenged training sessions for some seniors.

The tablets have an interface specifically designed to be intuitive and friendly for its users. It has a host of pre-loaded applications, including: games (chess, checkers, tetris, etc.), television, a newspaper reader, a PDF reader, Wikipedia, e-book application, an mp3 player, a calendar, an application to access the Uruguayan Social Security Bank's website, and links to different websites that include weather, pension information and bus schedules.

Volunteers taught seniors basic functions of the tablet that may help cognitive and non-cognitive stimulation –easy games, taking photographs-, connectedness with the family –applications for sending messages, video and audio calls-, contact with outside world –radio and newspapers webpages. The training includes three phases: 1) Individual training, 2) Group training, 3) Workshops, which were developed in that order.

- 1) Individual training was on a one to one basis, where each resident had a volunteer assigned, with whom they learned to perform the basic functions of the Tablet (firstly, understanding its operation and

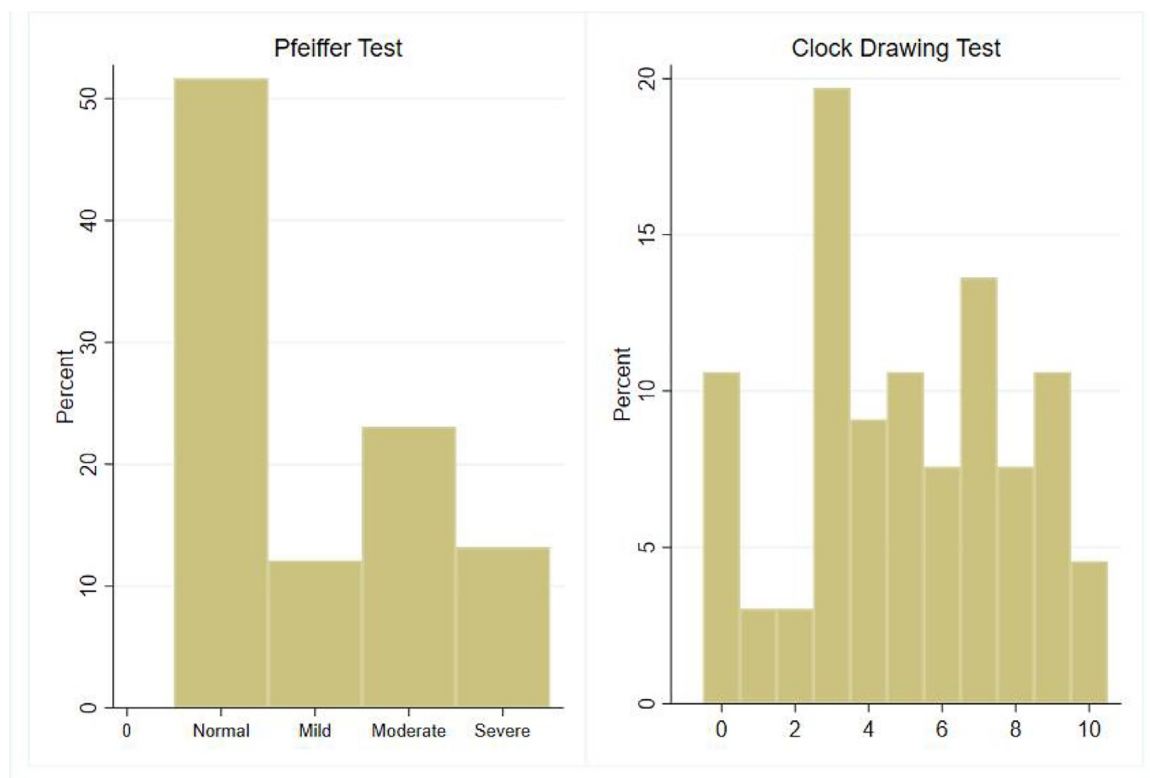


Fig. 2. Results of Clock drawing Test and Pfeiffer Test at baseline
Source: Administrative data provided by the authorities of the hospital.

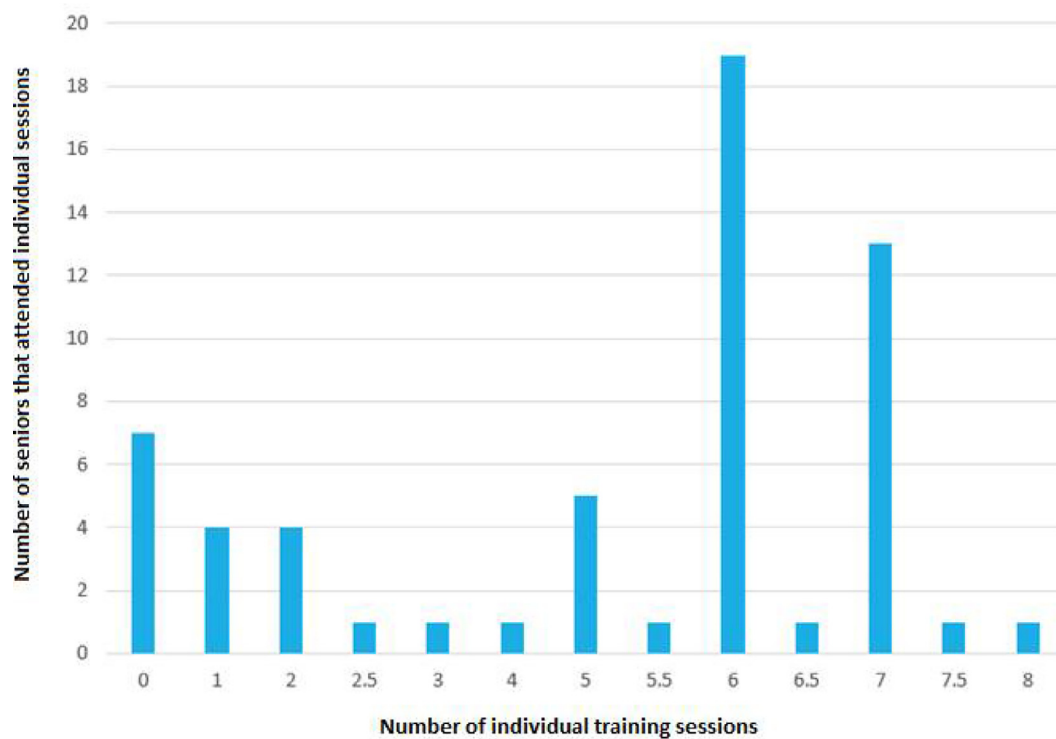


Fig. 3. Number of individual training sessions received by participants
Note: this figure offers the histogram of individual training sessions: the x-axis shows the number of individual training sessions (0.5 means incomplete session), and the y-axis reports the number of seniors that attend these sessions. Taller bars show that more seniors attended that number of sessions.

then evolving according to the personal preferences of the resident - certain functions, specific applications or areas of interest). It was planned to carry out seven trainings per resident. In the individual trainings, 57 residents and 53 volunteers participated. They worked for 2 months on individual training, completing 307 individual training sessions, with an average of 4.7 individual training sessions per resident selected to receive the Tablet. Fig. 3 summarizes the distribution of the individual training sessions received by participants. For instance, seven seniors of the treatment group did not attend individual classes; only one senior achieved to attend eight individual classes; most seniors attended six individual classes. Since some seniors failed to complete a specific training session, the value 0.5 was assigned to incomplete sessions. For example, Fig. 3 reports that one senior attended 2.5 sessions (i.e. two complete sessions plus an incomplete one).

- 2) Group training consisted of small groups - a maximum of three residents - trained by a volunteer. These sessions, carried out after individual training instances, had the objective of deepening what had been learned, advancing, fostering individual ownership of the tool, and generating autonomy for its use. It was planned to carry out seven sessions per group. In the group training, 57 residents and 30 volunteers participated. They worked under this modality for 3 months completing 140 sessions - an average of 3.6 group-training sessions per resident selected to receive the Tablet.
- 3) The workshops consisted of a weekly session of one hour in groups, carried out by more than one volunteer. The training was more relaxed, as it aimed to accompany the resident while still learning and using the Tablet. Five workshops were held per week. In the workshops, 47% of the elderly and 10 volunteers participated. In total, they worked for 17 weeks completing 85 workshops.

Most tablets were stored by the hospital's Secretariat and were recharged in this office. A number and name corresponding to the resident who worked with it identified each tablet. A few residents kept their tablets by themselves. In a typical day of training, the volunteers arrived at the hospital and passed through the Secretariat to take the tablet. Then the volunteer went to the ward where the resident with whom she had to work was. Training lasted one hour and a half in the case of individual or group trainings and one hour in the case of workshops. After completing the training, tablets were taken back to the Secretariat. In the absence of trainers -both during the program or after its completion-, participants were allowed to ask for a tablet in the Secretariat.

4.4. Identification strategies

This study employs two basic designs widely used in evaluating social programs: a randomized experimental design (also known as a randomized controlled trial) and a difference-in-differences (DD) design. In the randomized design, a group of elders, who satisfy the

eligibility requirements for the program, are randomly divided into two groups: the *treatment group*, who are assigned to receive the program, and the *control group*, who are assigned *not* to receive the program. Because assignment is random, treatment and control groups would be expected to have similar experiences in the post-program period in the absence of the intervention. Randomization therefore provides a simple method for constructing a counterfactual for the treatment group, using the observed outcomes of the control group. The estimate of the program effect from a randomized evaluation is simply the difference in post-program outcomes between the treatment group and the control group (Card et al., 2011).

The second methodological approach -DD- uses information from a non-randomly selected group of program nonparticipants - known as the *comparison group* - to construct the counterfactual outcomes of the participant group. In general, the observed difference in the post-program outcomes of the participant group and the comparison group will include both the true treatment effect and the selection bias component due to underlying differences between the two groups. The idea of a DD design is to compare the difference in outcomes between the participant group and the comparison group in some period *after* the participants have completed the program with the difference that existed *before* the program. Provided that the selection bias component is constant over time, the *change* in the difference between the participant group and the comparison group from before to after (i.e., the difference in differences) is an "unbiased" estimate of the effect of the program. The assumption in a DD design is that any pre-program differences between the participant and comparison groups would have remained fixed in the absence of the program (Card et al., 2011).

4.5. Randomized experimental design

A natural opportunity for introducing randomization occurred because the demand for the program (93 individuals identified by the authorities of the hospital) exceeded supply (the program delivered tablets and training for only 60 individuals). All 93 applicants were subject to an interview. In this baseline survey, we collected data on a wide array of elders' characteristics. From this population, 60 applicants were randomly assigned to the treatment group while the remaining candidates were assigned to the control group. The sequence in the process of randomization was designed to eliminate any likelihood of bias in group assignment. Randomization was done independently - the directors, nurses and supervisors of the hospital had no participation in any part of the randomization - and the sequence was concealed until the assignment occurred (the person enrolling participants did not know in advance if any elder would end in the treatment or control group).

A necessary condition for the validity of the impact evaluation results is that every pre-treatment characteristic must be balanced between the control group and the treatment group (the balancing condition). Table 3 reports that every variable is balanced in the baseline.

Table 3

Mean comparison of baseline characteristics.

Source: Survey applied at baseline by the research team and administrative data provided at baseline by the hospital authorities.

	Mean difference	P-values	Mean Control	Mean Treatment	Control #Obs.	Treatment #Obs.
Age (in years)	1.88	0.26	79.27	77.40	33	58
Male	0.10	0.35	0.48	0.38	33	60
Education (in years)	-0.87	0.27	5.21	6.08	33	59
Number of months in hospital	-10.15	0.48	53.15	63.30	33	60
Previous use of technology (1 = high; 0 = low)	-0.08	0.43	0.24	0.32	33	59
Dementia	0.05	0.64	0.48	0.43	33	60
Cognitive medication	0.01	0.89	0.21	0.20	33	60
Antidepressants medication	-0.02	0.89	0.48	0.50	33	60
Attend workshop	0.03	0.78	0.73	0.70	33	60
Subjects with External contact	0.02	0.88	0.58	0.56	33	59
Not living at the hospital (only daytime attendance)	-0.10	0.28	0.18	0.28	33	60

Since the variables age, education, previous use of technology, dementia and only day-time attendance suggest a better baseline condition –though not statistically significant– for the treatment group, it might be underlined as a limitation of the randomization process. Though we are not able to rule out completely this possibility, we should bear in mind that the mean of the other five variables suggest quite the opposite: the control group spent less time institutionalized, showed greater external contact, consumed less antidepressants, and attended workshops in a greater proportion at baseline.

In principle, randomization renders baseline surveys unnecessary, since it ensures the treatment and control groups are similar. However, there are some reasons why researchers may still want to conduct a baseline survey (Duflo et al., 2007). First, a baseline survey generates control variables that will reduce the variability in final outcomes and therefore may reduce sample size requirements. Second, it makes it possible to examine interactions between initial conditions and the impact of the program. Finally, a baseline survey provides an opportunity to check that the randomization was conducted appropriately and offers an opportunity to test and refine data collection procedures.

The primary purpose of this study is to determine the causal effect of participating in the tablet program on elders' cognitive deterioration and connectedness. Formally, we estimate the following equation:

$$Y_{is} = \alpha + \beta \cdot T_i + \theta'X_i + \mu_s + \varepsilon_{is} \quad (1)$$

where Y_{is} is any of the outcomes of interest (dementia, cognitive and antidepressant medication, workshop attendance and external contact, and values from four tests: Pfeiffer test, Clock drawing test, self-esteem test, loneliness test) for individual i and ward s ; T_i is a dummy variable that takes the value of one for individuals assigned to the treatment group and zero otherwise; X_i correspond to individual level controls (age, gender, number of years of schooling, previous use of technology, months in hospital); μ_s corresponds to fixed effects by ward s ; α is the intercept; β is the parameter of interest; and ε_i is the error term.

4.6. Difference in differences design

The integrity of a DD design hinges on whether it is really true that the pre-program difference between the participants and the comparison group would have persisted in the absence of the program. If so, the design is valid. If not, program impacts estimated from a DD design will be biased (Card et al., 2011). As the assignment for participation in the program came from an initial randomization, we may assume that both groups are designed without bias.

The simplest difference-in-differences estimate of the impact of the treatment (the use of tablets in the hospital) on cognitive and connectedness is estimated using the following regression equation:

$$Y_{ist} = \alpha + \beta \cdot M_{is} + \lambda \cdot d_t + \gamma \cdot (M_{is} \cdot d_t) + \theta'X_i + \mu_s + \varepsilon_{ist} \quad (2)$$

where Y_{ist} is the outcome reported by the individual i in ward s in the month t ; M_{is} is a dummy variable, taking a value of 1 if the individual i in ward s was assigned to the treatment group; d_t is a dummy variable that takes the value 1 for the follow-up moment; X_i correspond to individual level controls (age, gender, number of years of schooling, previous use of technology, months in hospital); μ_s corresponds to fixed effects by ward s ; and ε_{ist} is the error term that depends on the individual, the ward and the moment of observation. The coefficient of interest will be γ , the differential change for outcomes of individuals from the treatment group in comparison with the control group, in response to the tablet program.

5. Results

5.1. Results from the field experiment approach

In Tables 4–8 we present the main findings. The follow-up round was conducted 9 months after the start of the program. The assessment

included observational data gathered by the hospital (dementia, cognitive and antidepressant medication, workshop attendance and external contact), results of tests administered by the hospital (Pfeiffer test, Clock drawing test) and tests administered by volunteers (self-esteem test, loneliness test).

The Pfeiffer and Clock drawing tests, and surveys on usage of cognitive medication, antidepressant medication, dementia, external contact, and workshop attendance, were performed at the beginning and at the end of the intervention on participants of both groups. Loneliness and self-esteem questionnaires were applied at the end of the intervention to treatment and control groups. Fig. 4 shows the status of loneliness and self-esteem for all participants –i.e. seniors from both groups (treatment and control) taking as a whole. This figure is a graphical display of these outcomes at follow up and provide a picture about the vulnerability of the studied population. Each bar groups numbers into ranges. Taller bars show that more data falls in that range. As can be seen, the vast majority feel lonely, and nearly 30 percent report medium and low self-esteem. It should also be noted that what is called “high self-esteem” does not necessarily mean that patients are in an excellent state - they are simply out of risk of low self-esteem. For instance, Roth et al. (2008) find “extremely high response probability for all (self-esteem scale) items as a consequence of which self-esteem cannot be differentiated at the upper end of the range”. “Medium self-esteem” would mean that even if there are no serious problems it is considered worthy of improvement. Finally, “low self-esteem” expresses significant problems of self-esteem.

Tables 4–7 report the results for Eq. (1) including controls (age, gender, number of years of schooling, previous use of technology, months in hospital and fixed effects by ward). We find a lack of significant treatment effects in all the outcomes. The Constant gives a hint about the mean of the outcome in each model when all regressors set to zero (while in Table 4 the Constant is significantly different from zero, in the following tables the absence of its significance may be caused by the lack of statistical power). The results of the estimates of Eq. (1) without controls are similar, and for the sake of simplicity, we do not include them (they are available from the authors upon request).

5.2. Attrition

As is common in random evaluations in extremely underprivileged populations, by the end of the intervention both the treatment and control groups experienced attrition, with both groups suffering an approximate 45% reduction. There were many causes of attrition but in general, they were related to the extreme vulnerability of the population. Most of the losses were due to the inability to complete the questionnaires. Many patients were initially designated as capable of participation, but when questioned by the researchers were unable to answer, either because the patient was not coherent or did not understand many of the questions or concepts. Some of the day patients could not finish due to illness or having to remain at home due to medical indications. In addition, there were also a few deaths.

We compare the pre-treatment characteristics between those subjects who remained in the treated/control groups. Table 8 shows that all variables remain balanced.

Though the variables in the baseline remained balanced in the follow-up, the rate of attrition is too important to be ruled out as a potential source of bias. Thus, as a robustness check, we employed the DD design, exploiting the fact that the Pfeiffer test, the Clock drawing test, and the survey on usage of cognitive medication and on antidepressant medication were performed at the beginning and at the end of the intervention, to participants of both groups.

5.3. Results from the difference-in-difference approach

Previous research shows that frail and institutionalized older adults are particularly vulnerable to cognitive deterioration and isolation.

Table 4
Effects of Tablets on non-cognitive and cognitive aging and socioemotional outcomes (multivariate approach).

VARIABLES	Dementia	Cognitive medication	Antidepressant medication	Attend workshop	Subject with External Contact
Treatment	0.108 (0.0687)	0.108 (0.137)	−0.108 (0.139)	−0.0592 (0.112)	−0.0408 (0.105)
Constant	−1.154*** (0.402)	1.983** (0.797)	1.978** (0.823)	1.459** (0.668)	1.595** (0.638)
Observations	52	58	62	74	79
R-squared	0.807	0.148	0.107	0.172	0.227
Controls	Yes	Yes	Yes	Yes	Yes

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Notes: The estimates come from multivariate regressions employing OLS (Ordinary Least Squares). Standard errors in parentheses. The variable *treatment* equals 1 if the resident receives a Tablet. We display here the estimates of its coefficient β and also the Constant α (corresponding to Eq. (1)). Controls includes age, gender, number of years of schooling, previous use of technology, months in hospital and fixed effects by ward.

Table 5
Effects of Tablets on Pfeiffer test and Clock drawing test (multivariate approach).

VARIABLES	Pfeiffer test	Clock drawing test	Pathology Clock drawing test
Treatment	0.193 (0.805)	1.281 (1.154)	0.0166 (0.132)
Constant	−0.374 (4.668)	10.310 (6.974)	0.432 (0.824)
Observations	66	36	54
R-squared	0.313	0.258	0.162
Controls	Yes	Yes	Yes

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Notes: The estimates come from multivariate regressions employing OLS (Ordinary Least Squares). Standard errors in parentheses. The variable *treatment* equals 1 if the resident receives a Tablet. We display here the estimates of its coefficient β and also the Constant α (corresponding to Eq. (1)). Controls includes age, gender, number of years of schooling, previous use of technology, months in hospital and fixed effects by ward.

Table 6
Effects of tablets on loneliness (multivariate approach).

Variables	Emotional loneliness scale	Social loneliness scale	Total loneliness scale	Total loneliness scale index
Treatment	−0.306 (0.597)	−0.600 (0.538)	−0.906 (0.817)	−1.803 (1.614)
Constant	5.612 (3.808)	4.720 (3.431)	10.33* (5.212)	8.980 (10.30)
Observations	47	47	47	47
R-squared	0.052	0.112	0.119	0.118
Controls	Yes	Yes	Yes	Yes

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Notes: The estimates come from multivariate regressions employing OLS (Ordinary Least Squares). Standard errors in parentheses. The variable *treatment* equals 1 if the resident receives a Tablet. We display here the estimates of its coefficient β and also the Constant α (corresponding to Eq. (1)). Controls includes age, gender, number of years of schooling, previous use of technology, months in hospital and fixed effects by ward.

Theoretically, digital communication technologies, such as tablets, can contribute to foster cognitive skills and connectedness. Our study challenges this hypothesis. Tables 9 and 10 show the results of the differences in difference approach (Eq. (2)), that confirms the findings from the experimental approach (displayed in Section 5.1). We were unable to find any significant effect of the program.

Since some years ago, the hospital gathers regularly data on dementia, cognitive and antidepressant medication, workshop attendance, external contact, results of Pfeiffer and Clock drawing test. Thus, the directors of the hospital provide us with these administrative data –both in the baseline and follow-up. But the hospital does not gather data on self-esteem and loneliness, and we have no resources to collect

Table 7
Effects of Tablets on self-esteem (multivariate approach).

Variables	Self-esteem Score	Self-esteem score - Dichotomy
Treatment	−1.391 (1.133)	−0.810 (0.530)
Constant	7.429 (7.303)	−2.299 (3.415)
Observations	44	44
R-squared	0.383	0.313
Controls	Yes	Yes

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Notes: The estimates come from multivariate regressions employing OLS (Ordinary Least Squares). Standard errors in parentheses. The variable *treatment* equals 1 if the resident receives a Tablet. We display here the estimates of its coefficient β and also the Constant α (corresponding to Eq. (1)). Controls includes age, gender, number of years of schooling, previous use of technology, months in hospital and fixed effects by ward.

data on these scales at baseline. Hence, we were not able to assess the impact on these scales by the difference-in-difference approach. In future research, we should try to apply self-esteem and loneliness tests also at baseline. Anyway, the experimental approach, that does not need baseline data, reported no effect of the tablet program on self-esteem and loneliness.

6. Discussion

The main contribution of this study is that we are the first to evaluate the effect of an ICT intervention on cognitive deterioration and connectedness within a deeply vulnerable and institutionalized population. This paper describes the experience and results of the implementation of a one laptop per elderly plan (*Ibirapitá* Plan) in a public geriatric hospital, inhabited by a highly disadvantaged population, consisting of patients with dementia and low educational attainment, who are in general abandoned by their families, highly dependent, with many years of residency at the hospital and no previous usage of technology. The treatment included three phases: 1) Individual training, 2) Group training, 3) Workshops, which were developed in that order. The intervention was face-to-face and delivered by undergraduate students. They taught seniors basic functions of the tablet that may help cognitive and non-cognitive stimulation –easy games, taking photographs-, connectedness with the family –applications for managing photographs, sending messages, video and audio calls-, contact with outside world –radio and newspapers webpages. Firstly, students helped seniors to understand tablet's operations and then evolving according to the personal preferences of the resident. Group sessions and workshops had the objective of deepening what had been learned in the individual sessions, advancing, fostering individual ownership of the tool, and generating autonomy for its use.

The results from our analysis –employing an experimental and a

Table 8

Mean comparison of baseline characteristics at follow-up.

Source: Survey at baseline conducted by the research team & administrative data provided by the hospital about the 93 individuals selected by the hospital authorities to participate in the project.

	Mean difference	P-values	Mean Control	Mean Treatment	Control #Obs.	Treatment #Obs.
Age (in years)	-1.76	0.45	76.18	77.94	17	31
Male	0.02	0.89	0.35	0.33	17	33
Education (in years)	-0.55	0.63	5.82	6.38	17	32
Numbers of month in hospital	11.41	0.55	69.53	58.12	17	33
Previous use of technology (1 = high; 0 = low)	-0.17	0.25	0.29	0.47	17	32
Dementia	-0.03	0.75	0.12	0.15	17	33
Cognitive medication	0.06	0.49	0.12	0.06	17	33
Antidepressant medication	0.16	0.29	0.65	0.48	17	33
Attend workshop	-0.08	0.61	0.53	0.61	17	33
Subjects with external contact	0.02	0.88	0.65	0.63	17	32
Not living at the hospital (only daytime attendance)	-0.07	0.62	0.24	0.30	17	33

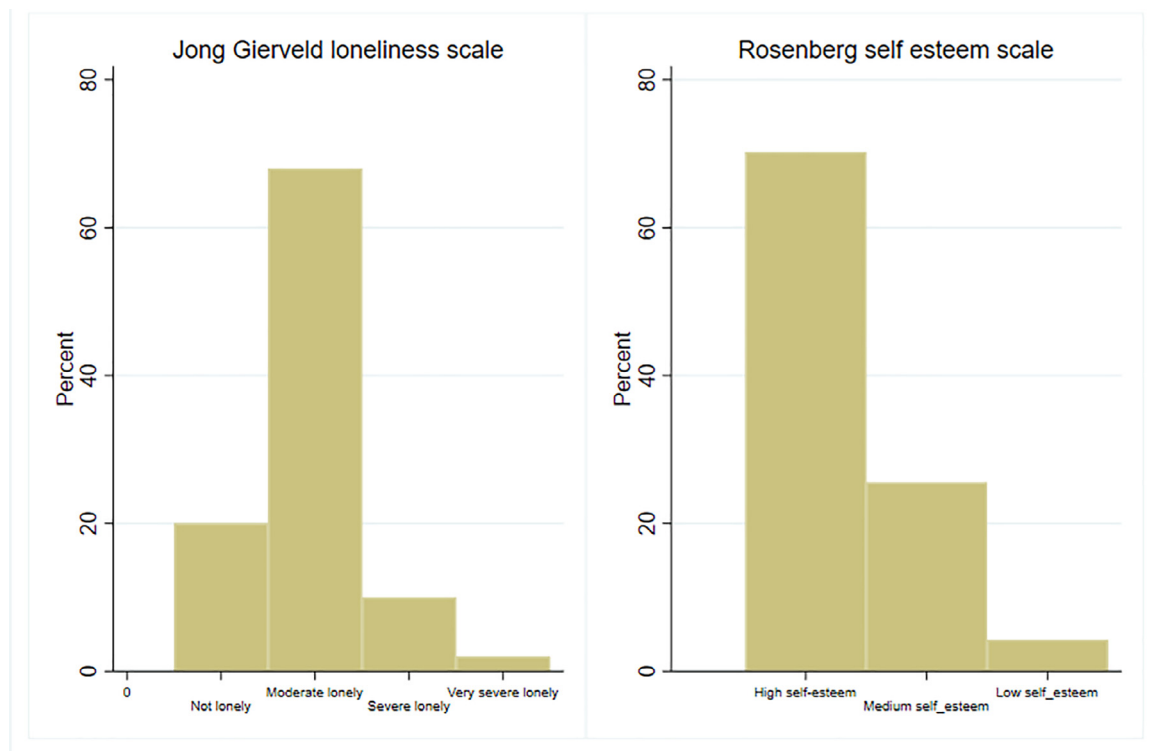


Fig. 4. De Jong Gierveld loneliness scale and the Rosenberg self-esteem scale at the follow-up

Source: Tests applied by the research team at follow-up.

difference-in-difference approach- about the effects on dementia, cognitive medication, antidepressant medication, external contact, participation in workshops, Pfeiffer test, and Clock drawing test show no significant differences between control and treatment group. We assessed also the effect on loneliness and self-esteem but only with an experimental strategy: we also find no impact.

The user group studied in our research does not represent all older adults in Uruguay. The conclusions of this study are limited to the special abilities and disabilities of the participants in the study, and cannot be generalized beyond them. It is quite possible that the broader elderly population can in fact benefit from the tablets. We have focused our research on the disadvantaged elderly because, compared with other senior groups, frail institutionalized older adults have received limited attention in the deployment and study of such technological programs.

The absence of effect of the tablet program is consistent with the qualitative ethnographic research of [Castleton et al. \(2020\)](#) and defies the techno-optimism among policy makers. These authors suggest that

public ICT interventions within vulnerable people may be challenged by the establishment of a sociotechnical network rife with complexities, disruptions, and fissures, and, last (but not least), by the fact that the user was heavily influenced by her self-perception of the potentialities of modern ICTs. Our findings are also consistent with [Duarte and Coelho \(2019\)](#). These authors suggest that some failures to address the needs of the elderly by technology are a consequence of poorly designed solutions. Duarte and his colleague designed different interfaces for a worldwide social network aiming to address these issues.

In the case of the tablet program studied in the current research, we were able to find many hints in the vein of [Castleton et al. \(2020\)](#) and [Duarte and Coelho \(2019\)](#). The tablets at the Piñeyro del Campo Hospital were connected to the internet through Wi-Fi. However, Wi-Fi coverage suffered from many problems. It was not present in all locations. Some wards were only partially covered, there were frequent periods of absence of internet availability due to the provider, lead to discouragement, frustration and loss of interest by the participants.

The free use of the tablets was restricted not only by internet

Table 9

Effects of Tablets on non-cognitive and cognitive aging and socioemotional outcomes (Differences in Differences approach).

VARIABLES	Dementia	Cognitive Medication	Antidepressant medication	Attend workshop	In touch with relatives
Treatment	0.0542 (0.0862)	0.0108 (0.0929)	−0.0258 (0.110)	−0.0553 (0.0934)	−0.0790 (0.0959)
Follow-up	−0.0341 (0.0430)	0.0175 (0.0721)	−0.0250 (0.0597)	−0.0846 (0.109)	0.0360 (0.0500)
Treatment x Follow-up	0.0740 (0.0555)	0.133 (0.0901)	0.0243 (0.0752)	0.0361 (0.137)	0.0308 (0.0634)
Constant	−0.353 (0.512)	0.810 (0.528)	1.649** (0.644)	0.982** (0.427)	1.351** (0.556)
Observations	142	148	152	164	169
Controls	Yes	Yes	Yes	Yes	Yes

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Notes: Standard errors in parentheses. We display here the estimates of Eq. (2). The dummy variable *treatment* takes the value 1 if the person has been selected to receive a Tablet. The dummy variable *follow-up* takes the value 1 for the observation in the follow up. The variable *treatment x follow-up* is a dummy variable that collects the interaction between *treatment* and *follow-up*: it takes the value 1 if the older adult was selected to receive a Tablet and the observation belongs to the follow-up moment. Controls include age, gender, number of years of schooling, previous use of technology, months in hospital and fixed effects by ward.

Table 10

Effects of Tablets on Pfeiffer test and Clock drawing test (Differences in differences approach).

Variables	Pfeiffer test	Clock drawing test	Pathology Clock drawing test
Treatment	−0.170 (0.666)	−0.0643 (0.795)	0.144 (0.110)
Follow-up	−0.0891 (0.348)	−0.149 (0.704)	0.00265 (0.109)
Treatment x Follow-up	0.379 (0.440)	0.552 (0.885)	−0.0903 (0.136)
Constant	−1.896 (3.893)	8.434 (5.132)	−0.382 (0.614)
Observations	154	99	117
Controls	Yes	Yes	Yes

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Notes: Standard errors in parentheses. We display here the estimates of Eq. (2). The dummy variable *treatment* takes the value 1 if the person has been selected to receive a Tablet. The dummy variable *follow-up* takes the value 1 for the observation in the follow up. The variable *treatment x follow-up* is a dummy variable that collects the interaction between *treatment* and *follow-up*: it takes the value 1 if the older adult was selected to receive a Tablet and the observation belongs to the follow-up moment. Controls include age, gender, number of years of schooling, previous use of technology, months in hospital and fixed effects by ward.

interruptions, but also due to operative problems with the charge and supply of the tablets. Some of the participants preferred to use other devices, easier to use and more familiar to them, such as TV, Radio and mobile phones. Many residents imagined the tablets as windows to the outside world, but their main use was for gaming and not for information nor for communication.

Another source of frustration was the prejudice of “I will not be able to do it”. Subjects were frustrated at not understanding the functioning of the tablets. Perseverance of both subjects and volunteers was shown to be of maximum importance.

For further research, it is important to assess the relative impact of the different features that shape a tablet intervention for the elderly: the contents of the training sections and the fidelity to these contents by the volunteers; the frequency, intensity and length of the intervention; the

training of the facilitators. In order to identify causality between tablets and seniors' outcomes (both cognitive and non-cognitive), it would be mandatory to design a field experiment that lets us disentangle the effect of having the device from the effect of having an instructor that assist the senior.

Heterogeneity of patients also plays an important role. There was a great diversity in the health status of the patients. Some were autonomous (e.g. they could walk go to the supermarket) while others were permanently prostrate in bed. Some were eager to learn, while others were fearful of their capability to learn, whilst others interrupted the training sessions giving various excuses.

Finally, we found that at the beginning of the intervention participants were very excited about receiving the tablet. However, anecdotal evidence suggests that it was the company of the volunteers during training, rather than the use of the tablets, that caused the enthusiasm. Castleton et al. (2020) deployed a qualitative ethnographic research on 15 seniors that participated in the tablet project at *Piñeyro del Campo* Hospital, comprised of semi-structured interviews, jottings, and field notes. The seniors they interviewed appreciated the tablet more by its perceived potential or the volunteers involved rather than by the tablet actual use:

‘As of now, [the tablet] has not been useful. It is if I get help from somebody else who knows. I can't communicate.’

‘I would try it, but as I tell you, I don't want to waste your time and the other kid's who are so good to me [referring to the volunteers].’

‘I love how you [volunteer] teach, but the problem is that I left it aside because as the thing to connect to the internet popped up and I don't know how to connect.’

(extracts from interviewed seniors; source: Castleton et al., 2020).

Disentangling the pure effect of tablets from the complementary presence of the volunteers deserves further research: it is key to explore these mechanisms to guide the assignation of resources in ICT interventions.

7. Conclusions

In this article, we present findings from an impact evaluation of a computer tablet program that took place in a long-term care facility for the elderly, in a country that experiences one of the greatest rate of aging population in Latin America. Trained volunteers met weekly, for approximately nine months, with seniors in a geriatric hospital to help them use the tablets. The project aimed at understanding the impact of the tablets on seniors' lives. Rigorous evaluations of ICT interventions for the elderly in deeply underprivileged populations were nonexistent in developing countries, yet, given the social, economic, and cultural differences, especially with respect to welfare systems, it cannot be assumed that the findings from the seminal studies in developed countries can be replicated.

In this study, we employed both a randomized controlled trial and a difference-in-differences design as identification strategies for the effects of the ICT intervention. Overall, we find no evidence of significant treatment effects in different dimensions of cognitive capabilities and connectedness of the tested elderly. It is unclear how malleable these dimensions are to short interventions given the mixed effects demonstrated in previous literature. Capabilities of deeply disadvantaged populations might be extremely difficult to change. Researchers and policy makers should avoid techno-optimism about the impact of technology on hospitalized seniors in their daily lives, and ought to take into account the contextual negotiations (access to Wi-Fi, the role of volunteers, the nurses, hospital facilities, etc.) that seniors engage with while using computers. These understudied issues deserve further research.

CRediT authorship contribution statement

Alejandro Cid: Conceptualization, Methodology, Writing - original draft, Writing - review & editing, Investigation. **Rafael Sotelo:** Writing - original draft, Visualization, Investigation. **Mariana Leguisamo:** Software, Data curation, Visualization, Formal analysis, Investigation. **María Ramírez-Michelena:** Project administration, Software, Investigation.

Declaration of Competing Interest

The Author declares that there is no conflict of interest.

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All the review was performed following the ethical standards.

Informed consent

All individual participants provided informed consent.

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